

The ordinary least squares solution

Readings for today

- Chapter 3: Linear regression. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). An introduction to statistical learning: with applications in R (Vol. 6). New York: Springer.

Topics

1. Maximum likelihood estimation

2. Model evaluation

Maximum likelihood estimation

Structure of a linear regression model

Fundamental form:

$$Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \epsilon$$

Solution:

$$\hat{\beta}_0 = E[Y] - \hat{\beta}_1 E[X]$$

$$= \bar{y} - \sum_{j=1}^p \hat{\beta}_j \bar{x}_j$$

$$\hat{\beta}_j = \frac{\sum_{i=1}^n (x_{ij} - \bar{x}_j)(y_i - \bar{y})}{\sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}$$

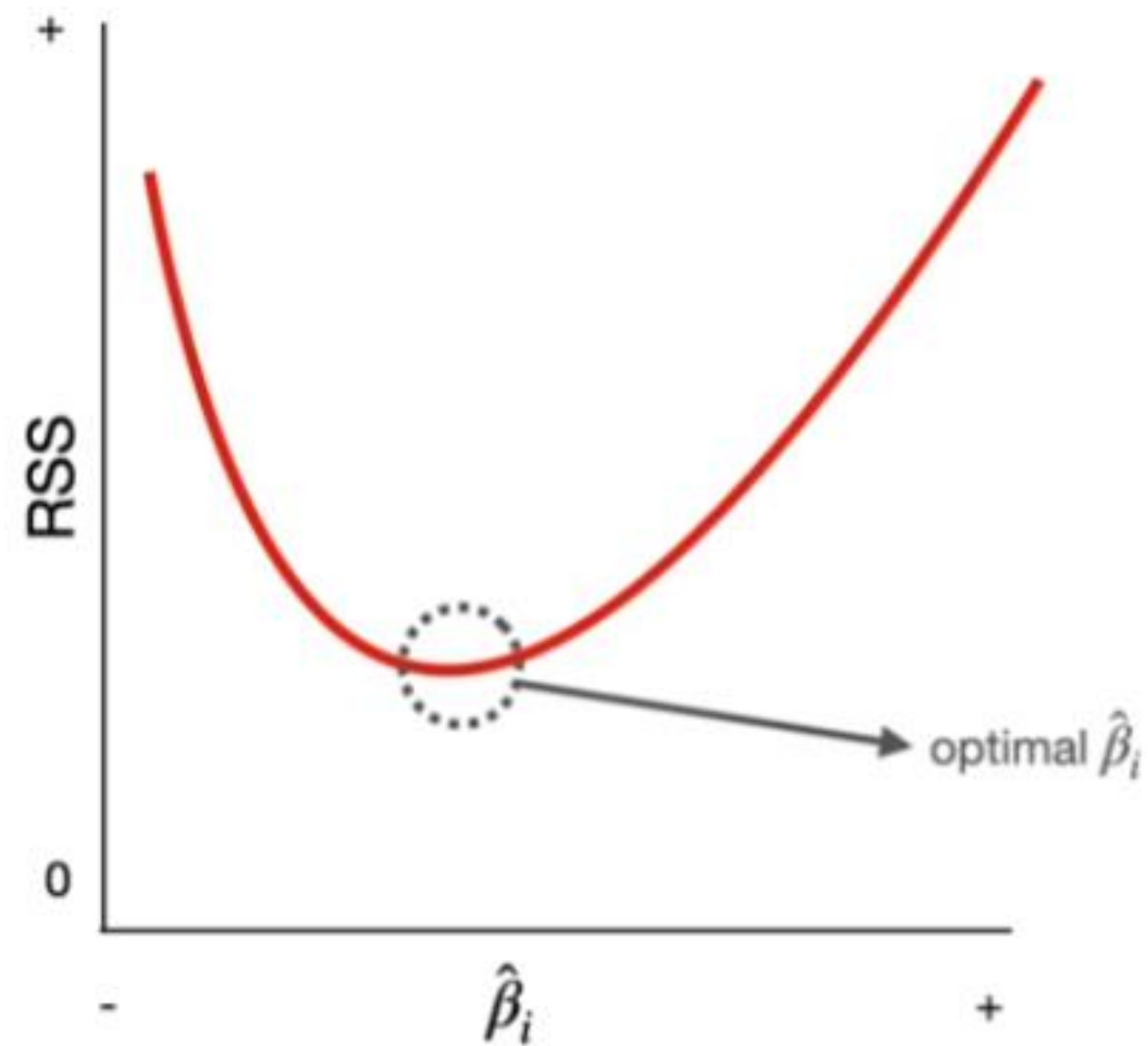
Necessary Assumptions to compute Ordinary Least Squares (OLS):

1. **Linear model specification:** predictors enter linearly in the parameters ($Y = X\beta + \epsilon$)
2. **Enough observations** ($n \geq p$): otherwise the columns of X cannot be linearly independent.
3. **No perfect multicollinearity:** No predictor is an exact linear combination of others.

+ Residuals of Y are normally distributed

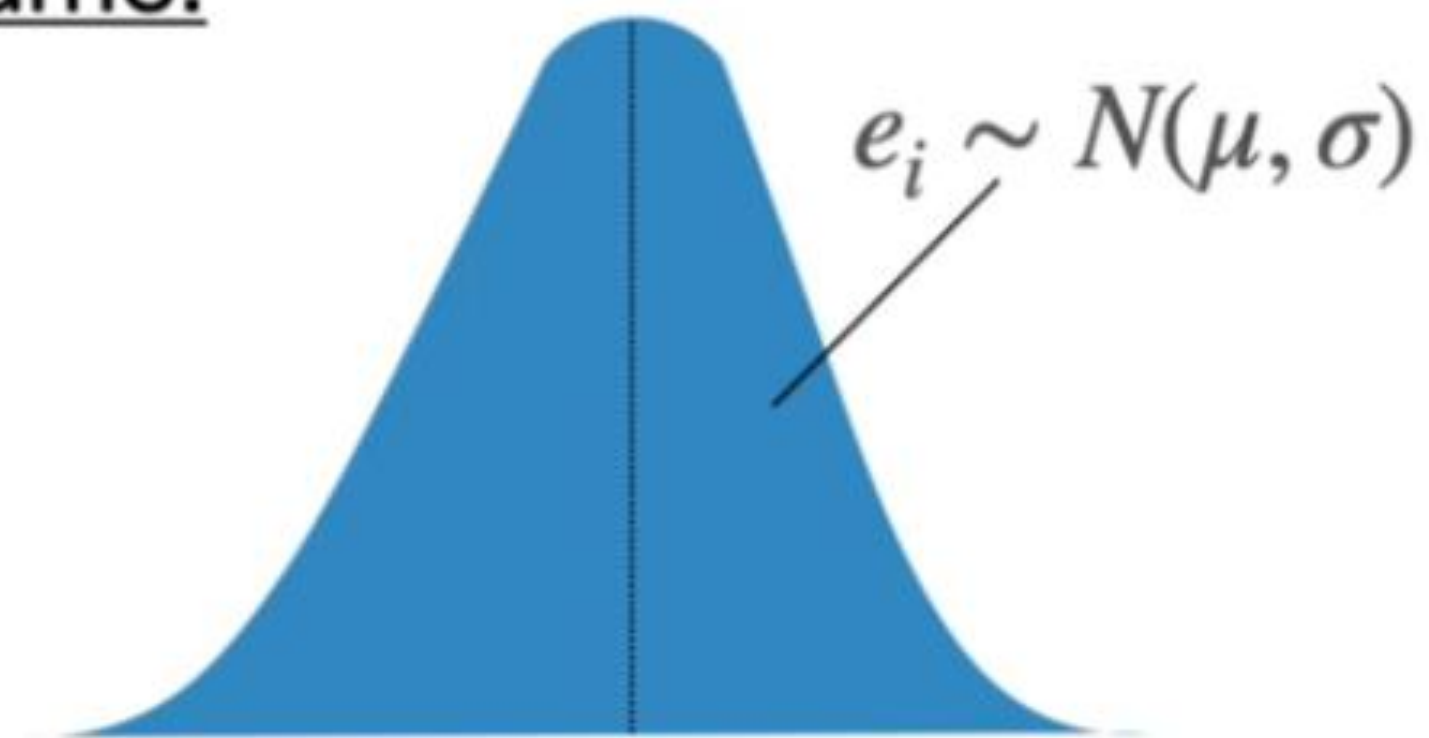
+ For every value x_i there is only one y_i

Residuals

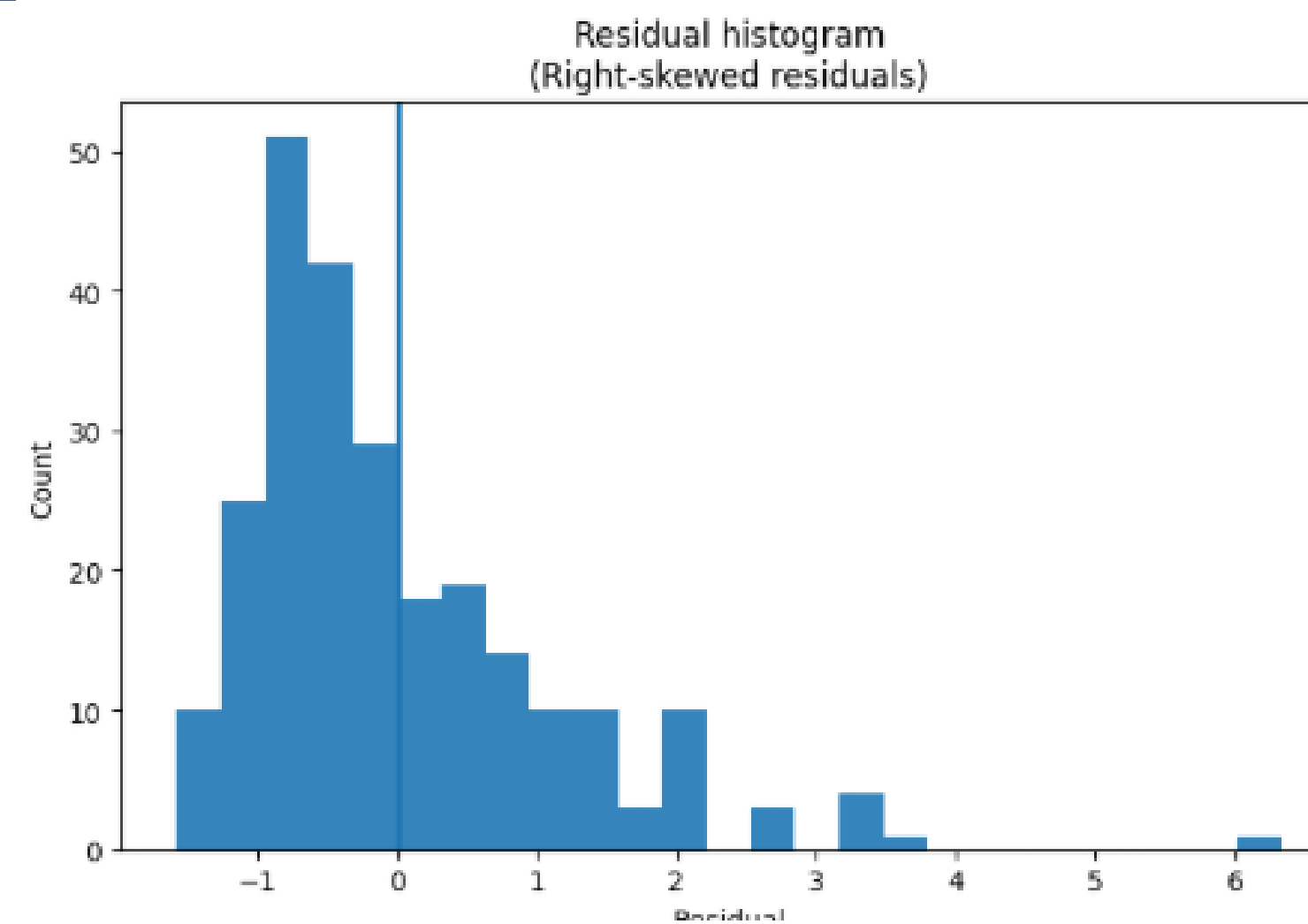
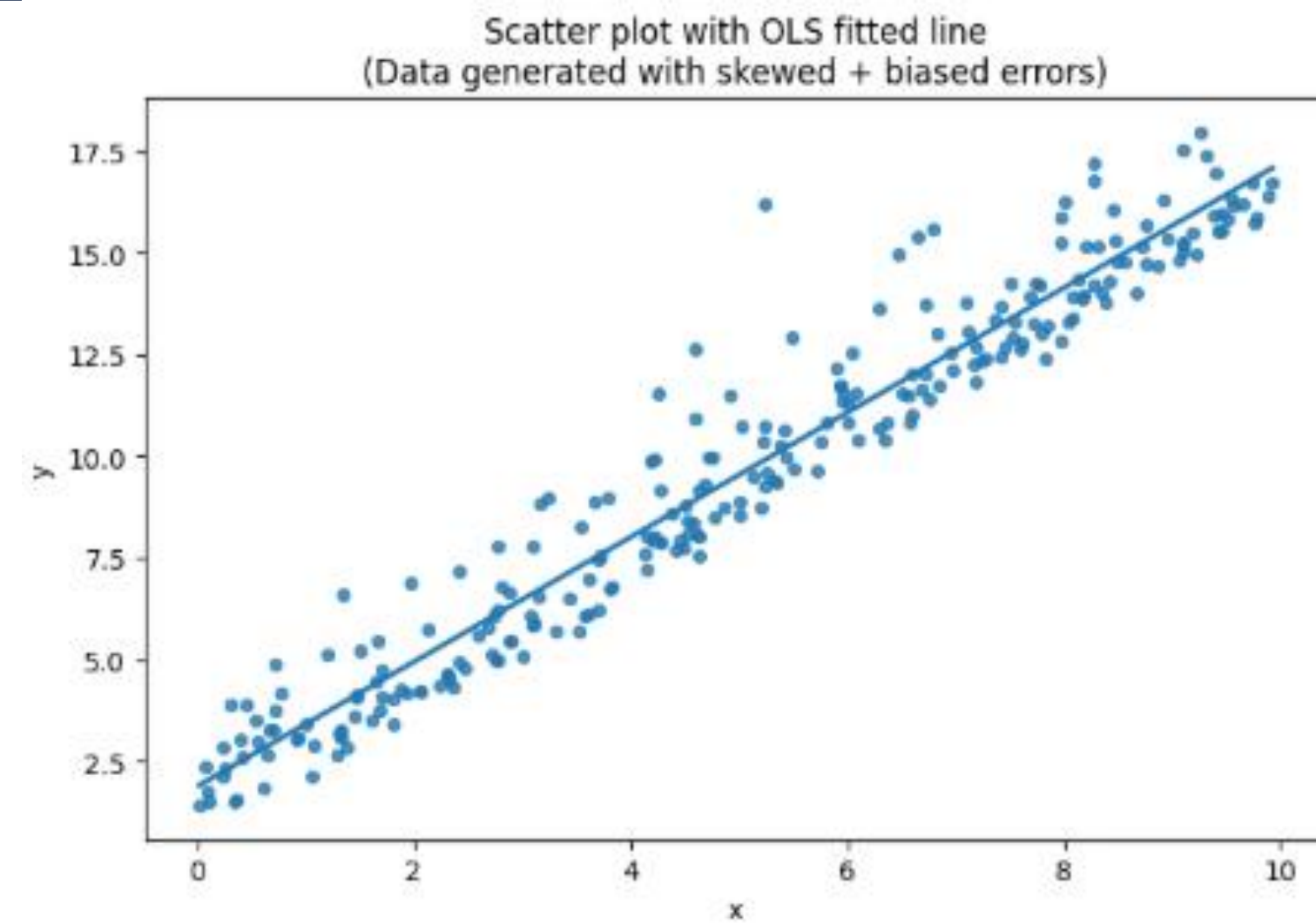


residuals: $(e_1^2, \dots, e_n^2)$
: $((y_1 - \hat{f}(x_1))^2, \dots, (y_n - \hat{f}(x_n))^2)$

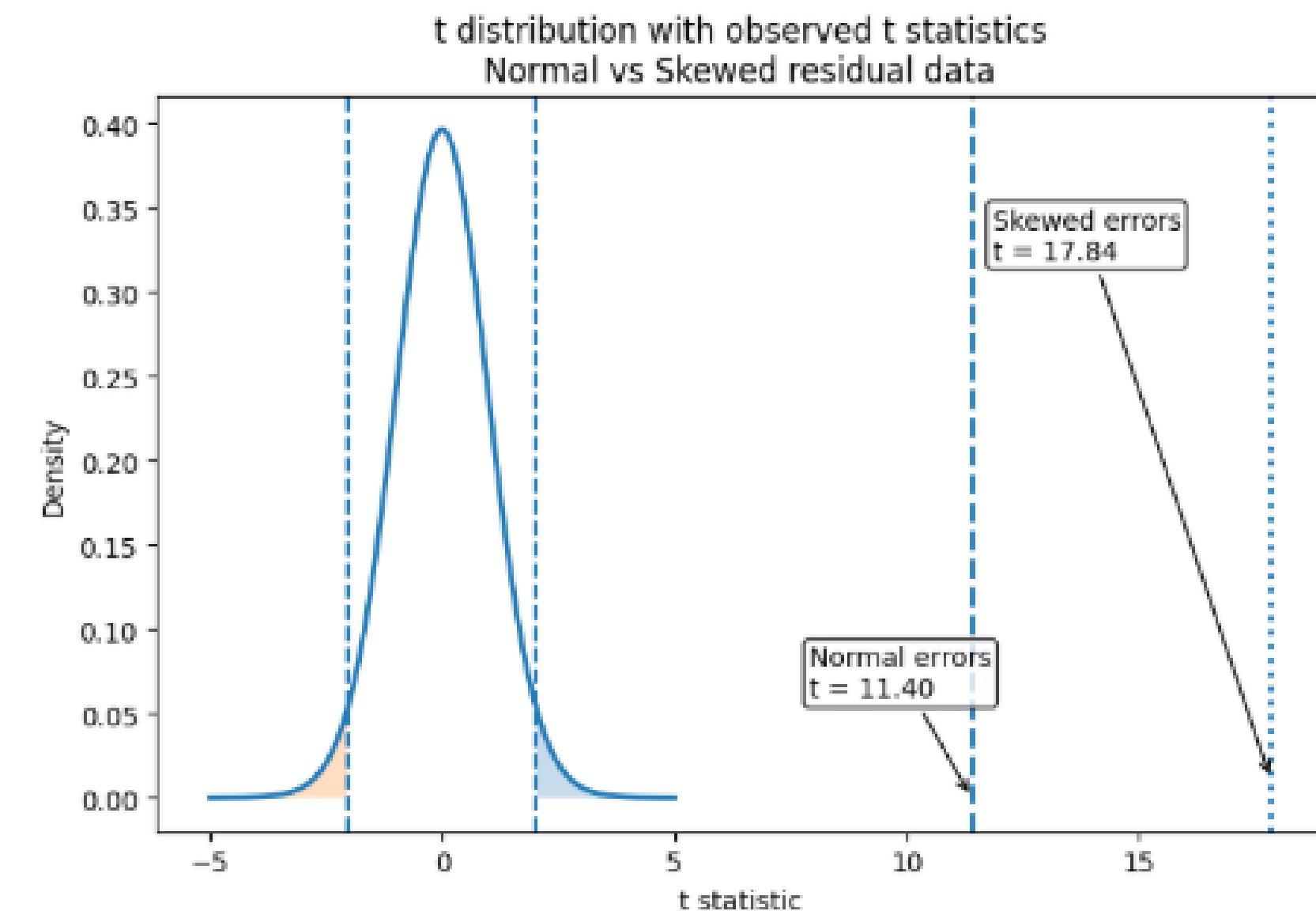
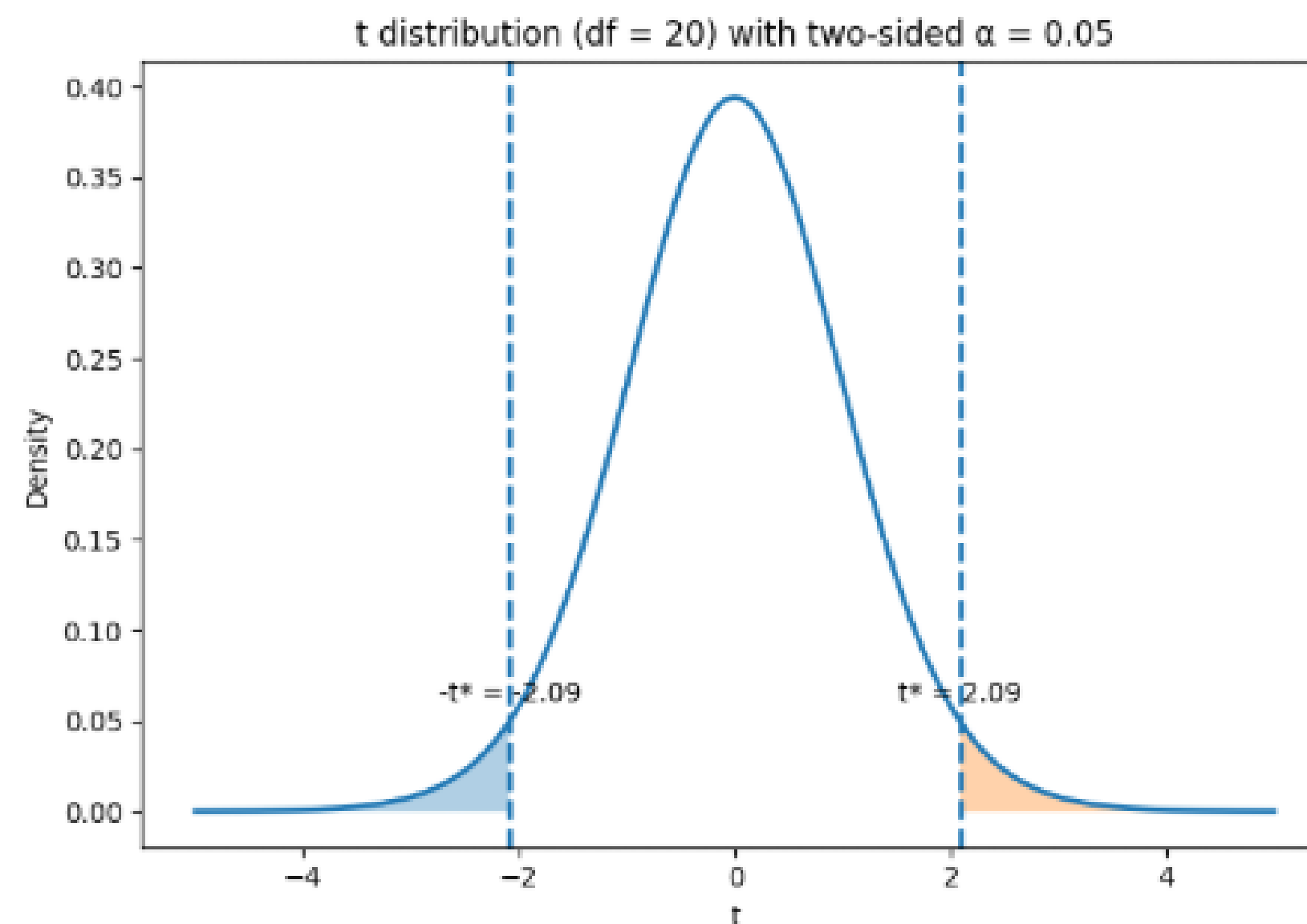
Assume:



Residuals



$$t = \frac{\hat{\beta}}{SE(\hat{\beta})}$$

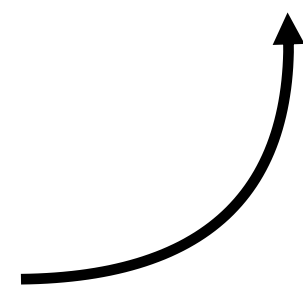


Likelihood function

The probability that the data you observe arises from a specific probability distribution with a specific set of parameters.

$$f(x|\mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Gaussian likelihood:



Returning to residuals

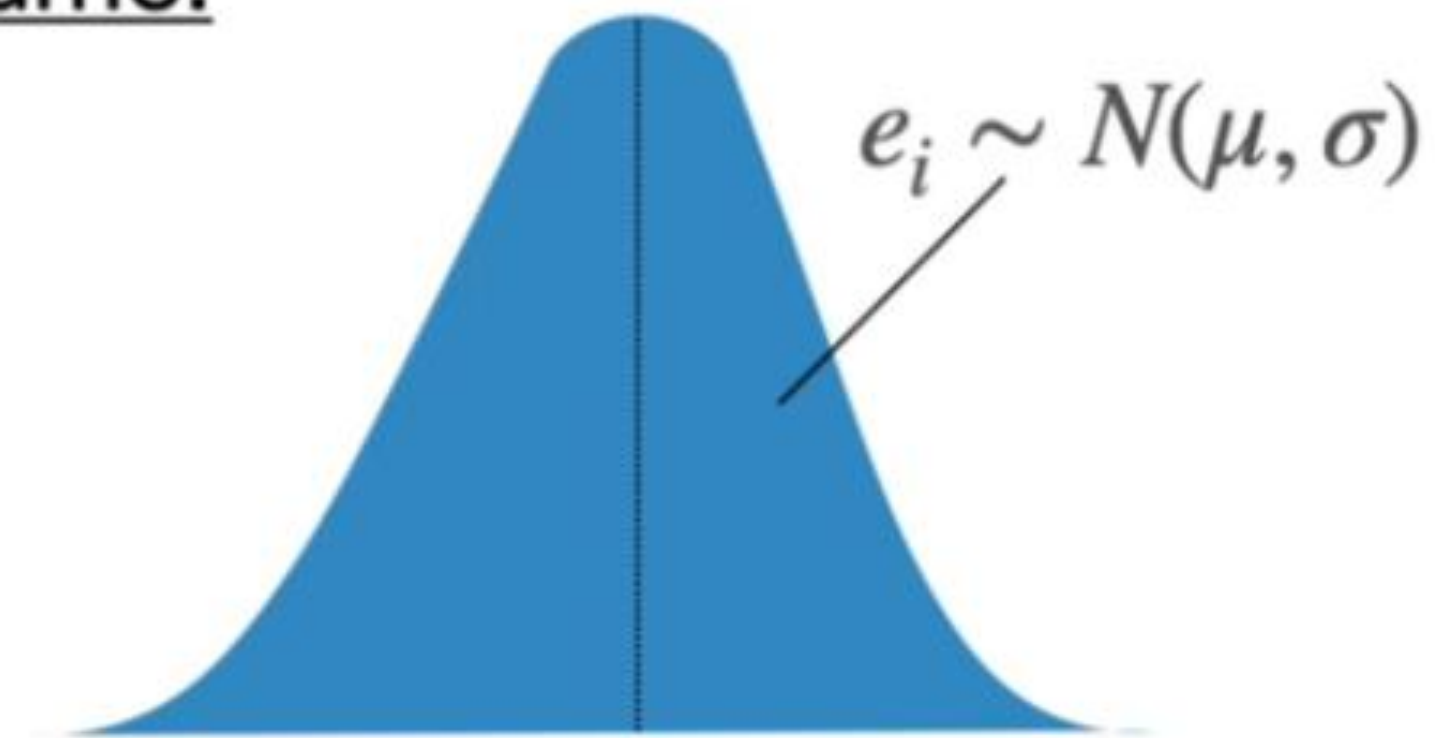
Likelihood of residuals:

$$L(\hat{\beta}_0, \dots, \hat{\beta}_p, \sigma) = \prod_{i=1}^n p(y_i | x_i; \hat{\beta}_0, \dots, \hat{\beta}_p, \sigma)$$

product $\nearrow$

$$= \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(y_i - \sum_{j=0}^p \hat{\beta}_j x_{ij})^2}{2\sigma^2}}$$

Assume:



Linear regression does not assume that Y is normally distributed.

It assumes that the deviations of Y around the regression line (the errors) are normally distributed *conditional on X* .

Refresher on logs

Just a reminder: $\log(e^{a-b}) = \log(e^a) - \log(e^b)$
 $= a - b$

Log likelihood of residuals

Likelihood of residuals:

$$L(\hat{\beta}_0, \dots, \hat{\beta}_p, \sigma) = \prod_{i=1}^n p(y_i | x_i; \hat{\beta}_0, \dots, \hat{\beta}_p, \sigma)$$
$$= \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(y_i - \sum_{j=0}^p \hat{\beta}_j x_{i,j})^2}{2\sigma^2}}$$

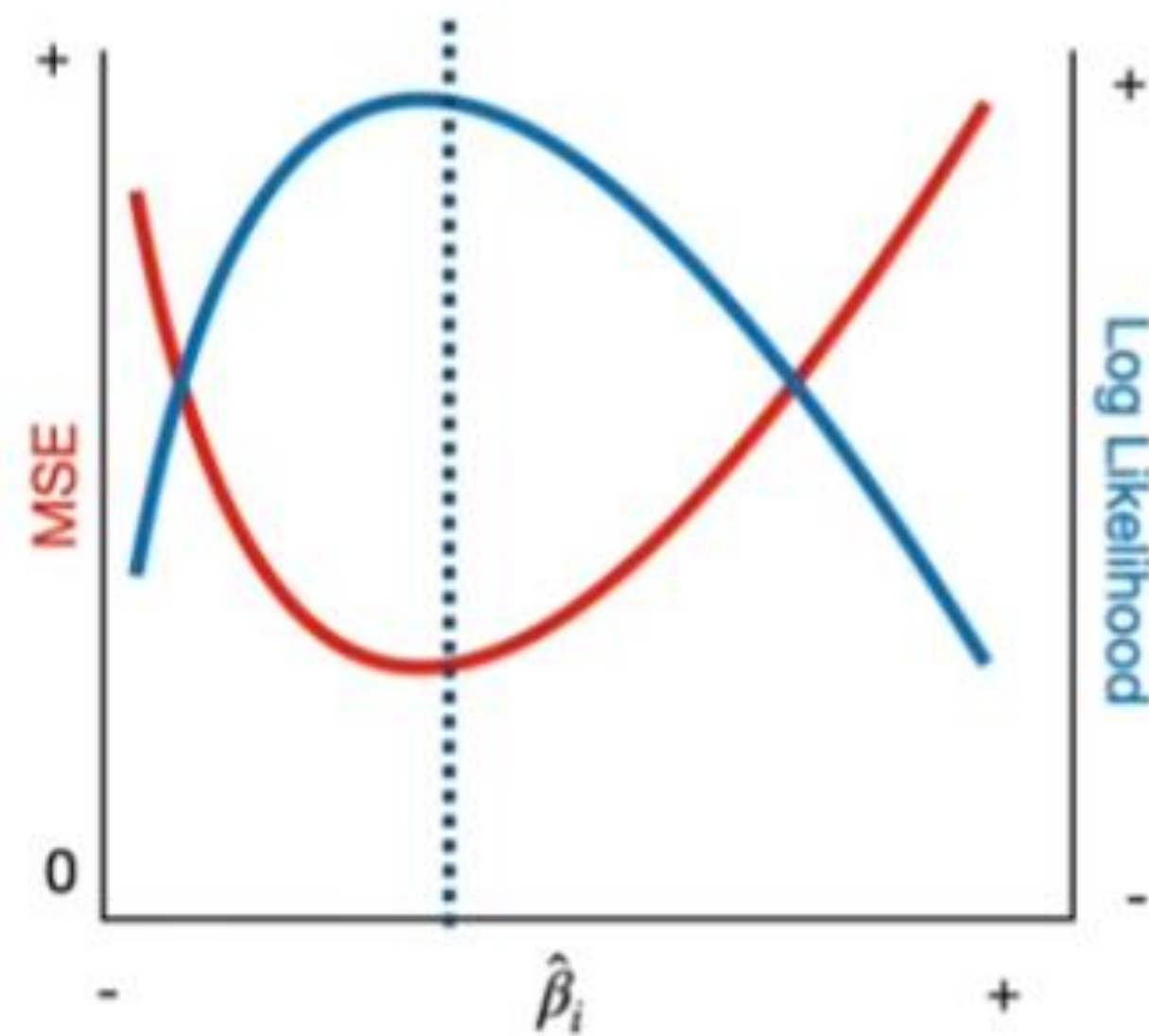
Log likelihood of residuals:


$$\ln(L(\hat{\beta}_0, \dots, \hat{\beta}_p, \sigma)) = \ln\left(\prod_{i=1}^n p(y_i | x_i; \hat{\beta}_0, \dots, \hat{\beta}_p, \sigma)\right)$$
$$= \sum_{i=1}^n \ln(p(y_i | x_i; \hat{\beta}_0, \dots, \hat{\beta}_p, \sigma))$$

$$= -\frac{n}{2} \ln(2\pi) - n \ln(\sigma) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \sum_{j=0}^p \hat{\beta}_j x_{i,j})^2$$

Maximum likelihood estimation

Maximizing the likelihood of your data given a particular model means finding the parameters that minimize your error (cost) function.



Task: Solve for $\hat{\beta}_0$ and any $\hat{\beta}_i$ where $i > 0$.

Mean Squared Error (MSE):

$$\begin{aligned} MSE(\hat{\beta}_0, \hat{\beta}_1) &= E[(Y - (\hat{\beta}_0 + \hat{\beta}_1 X))^2] \\ &= E[Y^2] - 2\hat{\beta}_0 E[Y] - 2\hat{\beta}_1 E[XY] + E[(\hat{\beta}_0 + \hat{\beta}_1 X)^2] \\ &= E[Y^2] - 2\hat{\beta}_0 E[Y] - 2\hat{\beta}_1 (Cov[XY] + E[X]E[Y]) \\ &\quad + \hat{\beta}_0^2 + 2\hat{\beta}_0 \hat{\beta}_1 E[X] + \hat{\beta}_1^2 E[X^2] \\ &= E[Y^2] - 2\hat{\beta}_0 E[Y] - 2\hat{\beta}_1 Cov[XY] - 2\hat{\beta}_1 E[X]E[Y] + \hat{\beta}_0^2 \\ &\quad + 2\hat{\beta}_0 \hat{\beta}_1 E[X] + \hat{\beta}_1^2 Var[X] + \hat{\beta}_1^2 (E[X])^2 \end{aligned}$$

Tips & reminders

$$1. \quad \frac{\delta(x^2 + y)}{\delta x} = 2x$$

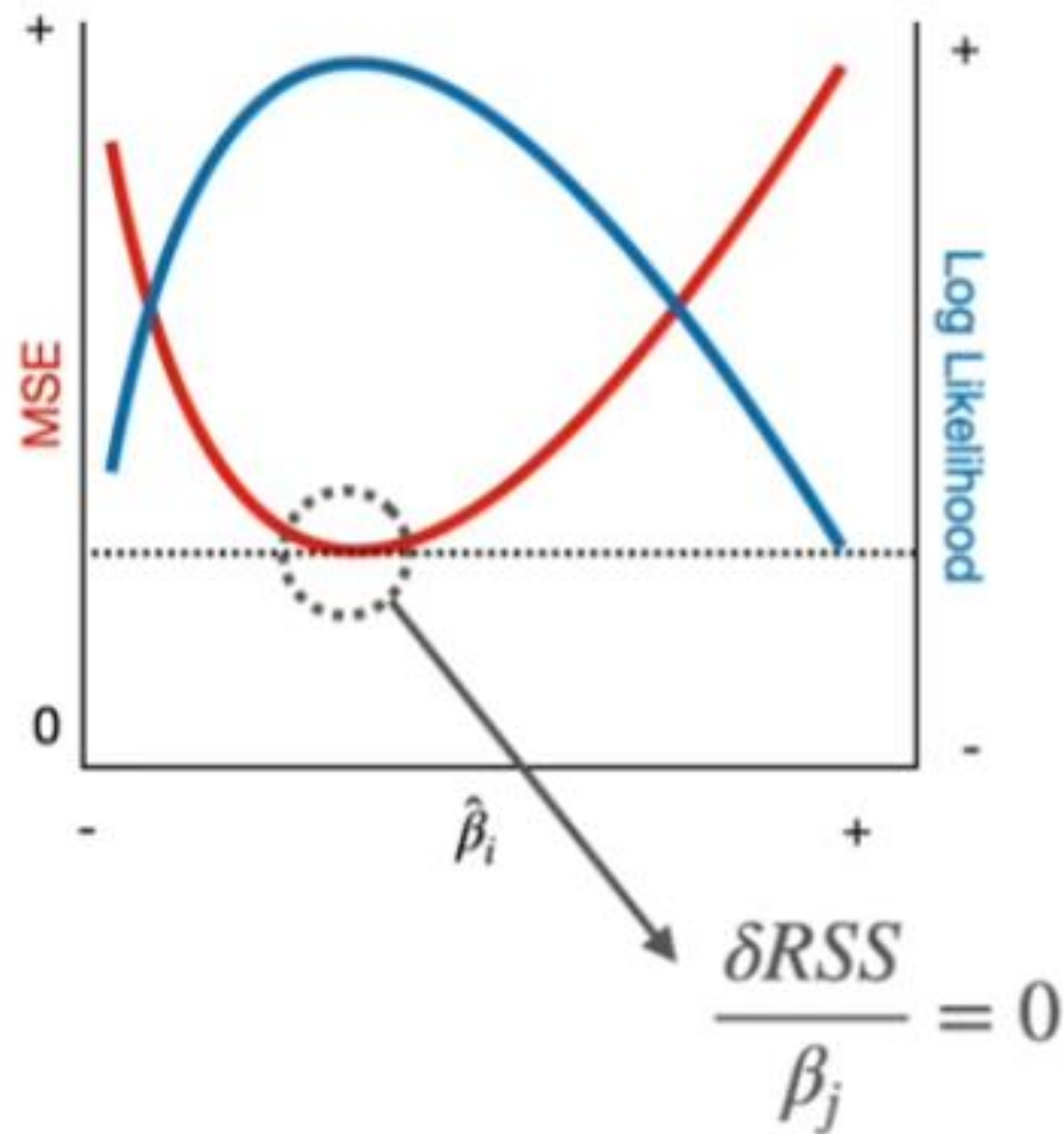
$$2. \quad E[X] = \bar{X}$$

$$3. \quad E[X^2] = Var[X] + E[X]^2$$

$$4. \quad E[XY] = Cov[X, Y] + E[X]E[Y]$$

Some useful reminders
from calculus & statistics
for solving the MLE for
ordinary least squares.

Maximum likelihood estimation

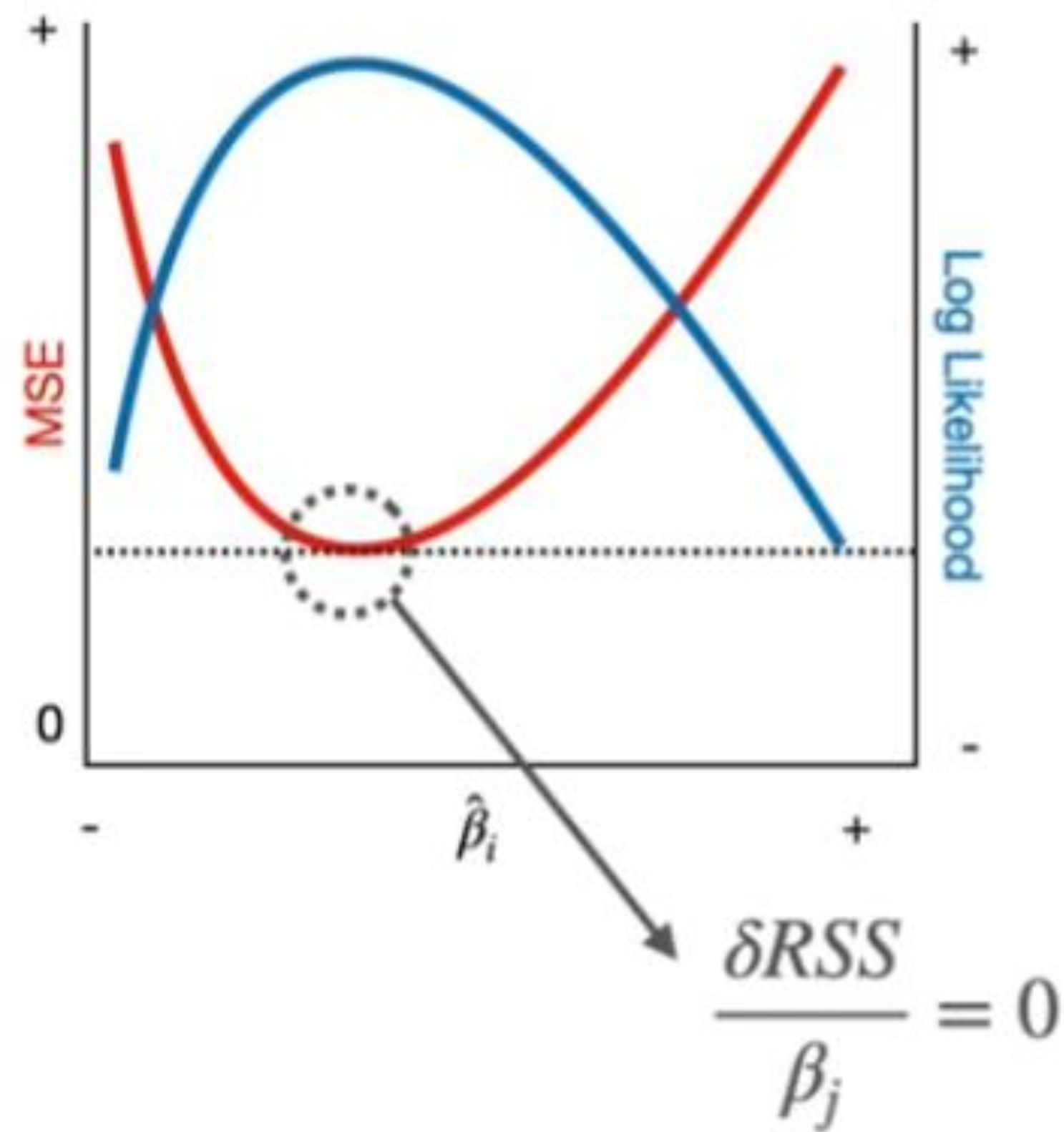


$$MSE(\hat{\beta}_0, \hat{\beta}_1) = E[Y^2] - 2\hat{\beta}_0 E[Y] - 2\hat{\beta}_1 Cov[XY] - 2\hat{\beta}_1 E[X]E[Y] \\ + \hat{\beta}_0^2 + 2\hat{\beta}_0\hat{\beta}_1 E[X] + \hat{\beta}_1^2 Var[X] + \hat{\beta}_1^2 (E[X])^2$$

$$\hat{\beta}_0: \frac{\partial E[(Y - (\hat{\beta}_0 + \hat{\beta}_1 X))^2]}{\partial \hat{\beta}_0} = -2E[Y] + 2\hat{\beta}_0 + 2\hat{\beta}_1 E[X]$$

$$\hat{\beta}_0 = E[Y] - \hat{\beta}_1 E[X] = \bar{y} - \hat{\beta}_1 \bar{x}$$

Maximum likelihood estimation

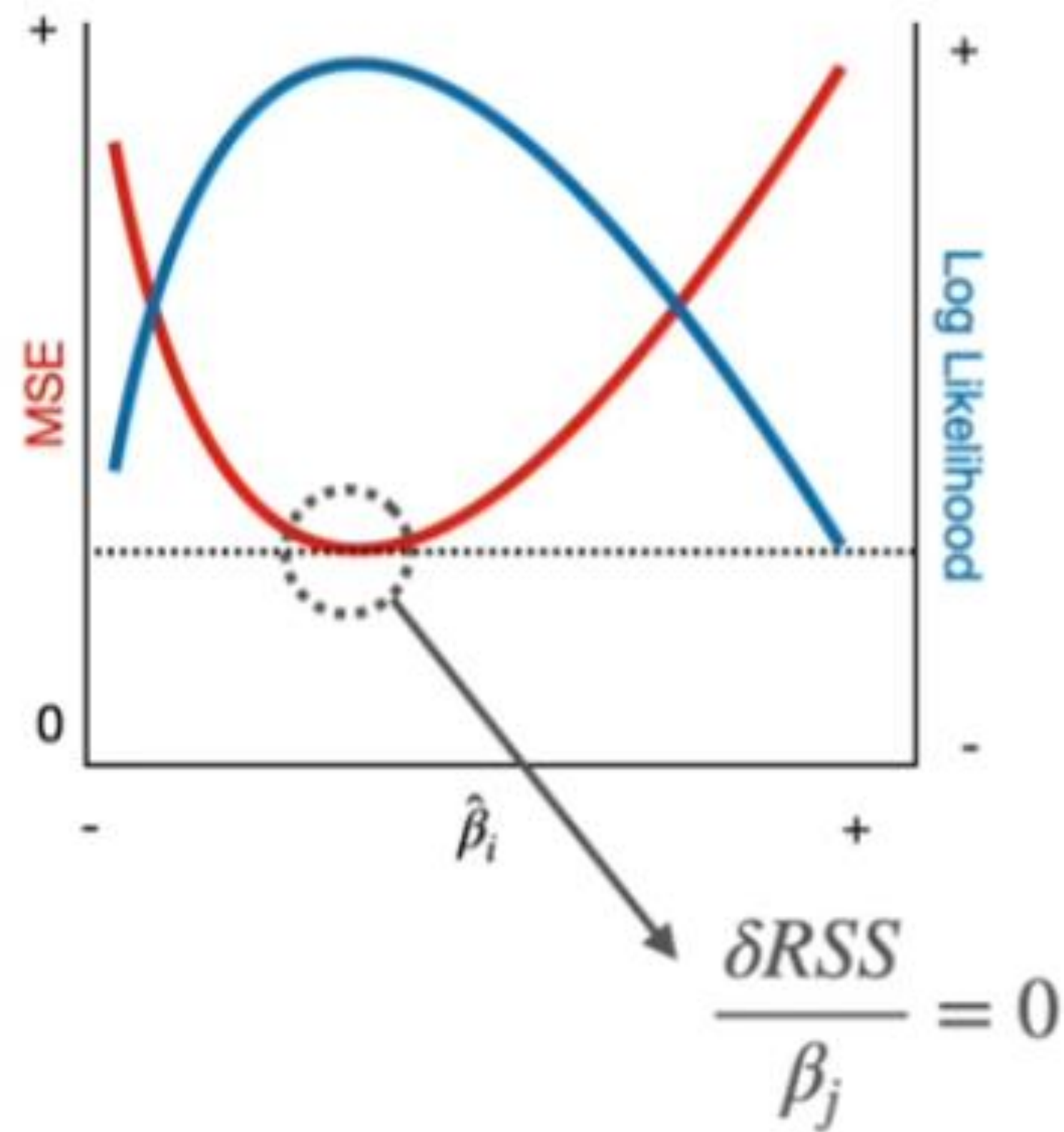


$$MSE(\hat{\beta}_0, \hat{\beta}_1) = E[Y^2] - 2\hat{\beta}_0 E[Y] - 2\hat{\beta}_1 Cov[XY] - 2\hat{\beta}_1 E[X]E[Y] \\ + \hat{\beta}_0^2 + 2\hat{\beta}_0\hat{\beta}_1 E[X] + \hat{\beta}_1^2 Var[X] + \hat{\beta}_1^2 (E[X])^2$$

$$\hat{\beta}_1: \frac{\partial E[(Y - (\hat{\beta}_0 + \hat{\beta}_1 X))^2]}{\partial \hat{\beta}_1} = -2Cov[XY] - 2E[X]E[Y] \\ + 2\hat{\beta}_0 E[X] + 2\hat{\beta}_1 Var[X] + 2\hat{\beta}_1 (E[X])^2$$

$$\hat{\beta}_1 = \frac{Cov[X, Y]}{Var[X]} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Maximum likelihood estimation



$$MSE(\hat{\beta}_0, \hat{\beta}_1) = E[Y^2] - 2\hat{\beta}_0 E[Y] - 2\hat{\beta}_1 Cov[XY] - 2\hat{\beta}_1 E[X]E[Y] \\ + \hat{\beta}_0^2 + 2\hat{\beta}_0\hat{\beta}_1 E[X] + \hat{\beta}_1^2 Var[X] + \hat{\beta}_1^2 (E[X])^2$$

$$\hat{\beta}_0 = E[Y] - \hat{\beta}_1 E[X] = \bar{y} - \hat{\beta}_1 \bar{x} \\ = \bar{y} - \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \bar{x}$$

Assumptions make sense

Necessary Assumptions to compute Ordinary Least Squares (OLS):

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+ Residuals of Y are normally distributed

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Advantages for MLE:

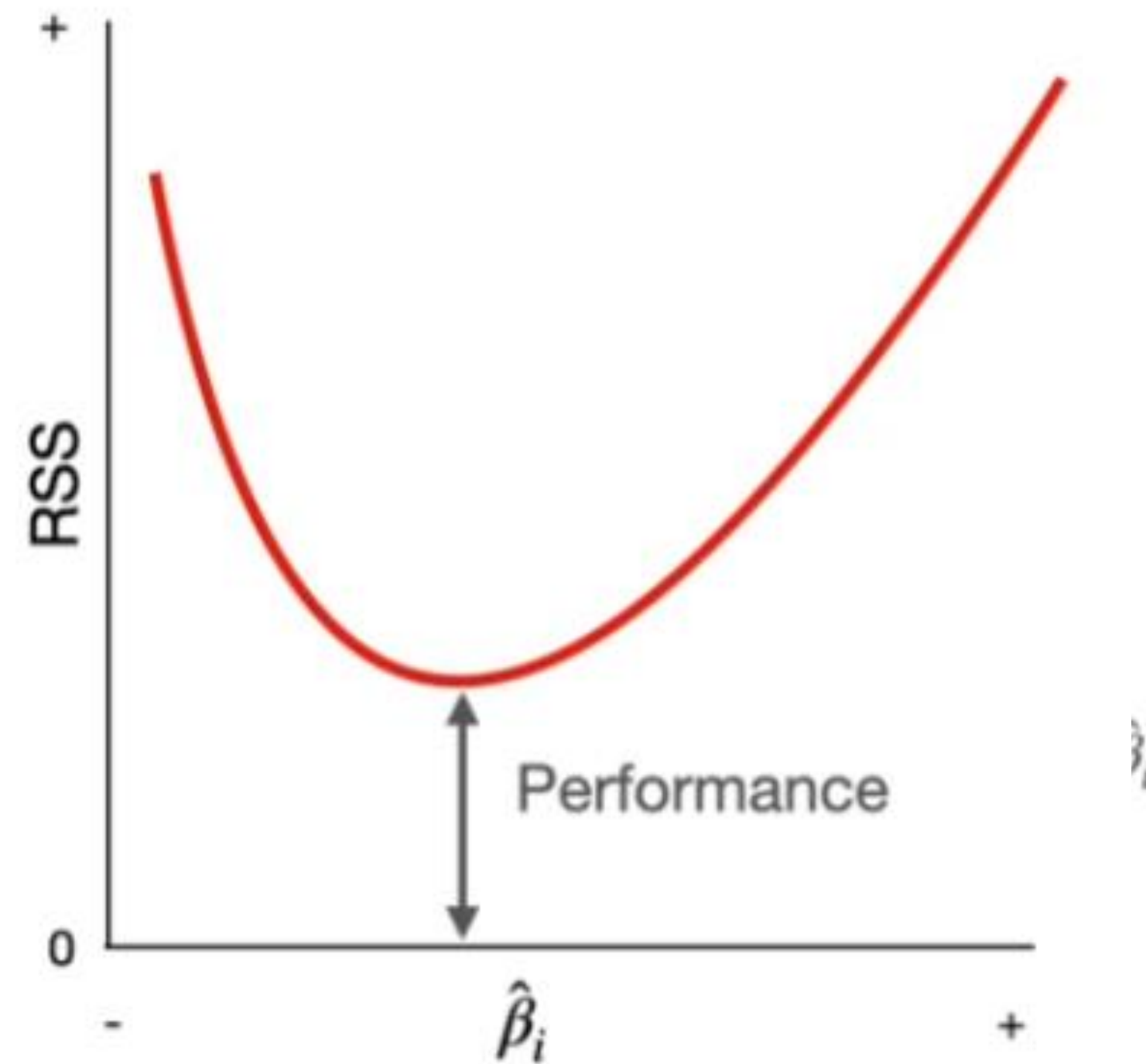
Well defined probability distribution & convex error gradient.

Allows for solving each $\hat{\beta}_i$ separately.

Only one function and it is solvable.

Model evaluation

Best you can do?



Ideal model: One that accounts for all of the variance in your data.

$$\begin{aligned} RSS &= e_1^2 + \dots + e_n^2 \\ &= \sum_{i=1}^n (y_i - \hat{\beta}_1 x_i)^2 \\ &= 0 \end{aligned}$$

Residual square error (RSE)

$$RSE = \sigma_{model}^2 = \sqrt{\frac{RSS}{n - k - 1}} = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n - k - 1}}$$

- Bounded at 0
- Simplest to calculate
- $\downarrow RSE = \uparrow model\ fit$
- Closes evaluation of $\hat{y}$ itself

Coefficient of determination (r^2)

$$r^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS} = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

- Direct measure of percent variance in Y explained by $\hat{f}$
- $\uparrow r^2 = \uparrow \text{model fit}$
- Meaningful units (0% $\rightarrow$ 100%)

F-test (F)

$$F = \frac{\frac{1}{p}(TSS - RSS)}{\frac{1}{n - p - 1}RSS} = \frac{\frac{1}{p}(\sum_{i=1}^n (y_i - \bar{y})^2 - \sum_{i=1}^n (y_i - \hat{y})^2)}{\frac{1}{n - p - 1} \sum_{i=1}^n (y_i - \hat{y})^2}$$

- Bounded at 0
- $\uparrow F = \uparrow$ *model fit*
- Ratio of variances (variance explained by $\hat{f}$ over variance of y)

Common theme

All model evaluation measures how well the prediction, $\hat{y}$, explains y against compared to the mean, $\bar{y}$.

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2 \ll \sum_{i=1}^n (y_i - \bar{y}_i)^2$$

RSS *TSS*

Fundamental form:

$$Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \epsilon$$

Take home message

- Ordinary least squares regression provides a simple, closed form solution to the model fitting problem... so long as the data meet the assumptions.